HARD

GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

06

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY

Which of the following equations represents a circle in the *xy*-plane that intersects the *y*-axis at exactly one point?

A. $x - 8^2 + y - 8^2 = 16$

B. $x - 8^2 + y - 4^2 = 16$

C. $x - 4^2 + y - 9^2 = 16$

D. $x^2 + y - 9^2 = 16$

Q2

GEOMETRY AND TRIGONOMETRY

Which of the following expressions is equivalent to $\sin 24° \cos 66° + \cos 24° \sin 66°$?

A. $2\cos 66° \sin 24°$

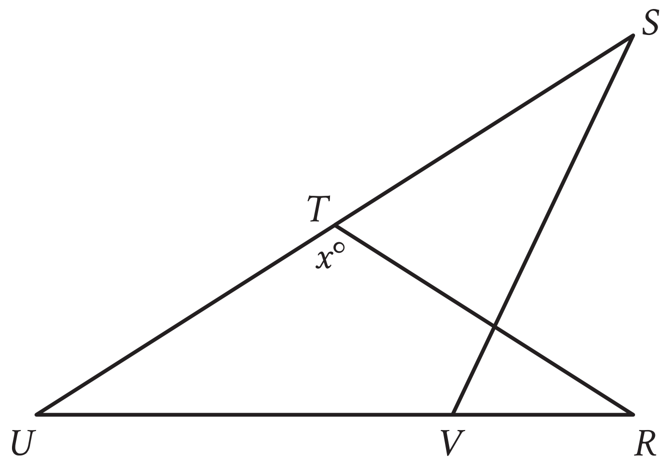
B. $2\cos 66° + 2\cos 24°$

C. $\cos 66°^2 + \cos 24°^2$

D. $\cos 66°^2 + \sin 24°^2$

Q3

GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Triangle upper R upper T upper U is labeled as follows:
    - The measure of angle upper R upper T upper U is x°.
- The 2 triangles partially overlap:
    - Side upper R upper T of triangle upper R upper T upper U intersects side upper S upper V of triangle upper S upper V upper U.
    - Vertex upper V of triangle upper S upper V upper U lies on side upper R upper U of triangle upper R upper T upper U.
    - Vertex upper T of triangle upper R upper T upper U lies on side upper S upper U of triangle upper S upper V upper U.
- A note indicates the figure is not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is $29°$, and the measure of angle RVS is $41°$. What is the value of x?

STUDENT-PRODUCED RESPONSE

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 GRID-IN GEOMETRY AND TRIGONOMETRY

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF. The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 GEOMETRY AND TRIGONOMETRY

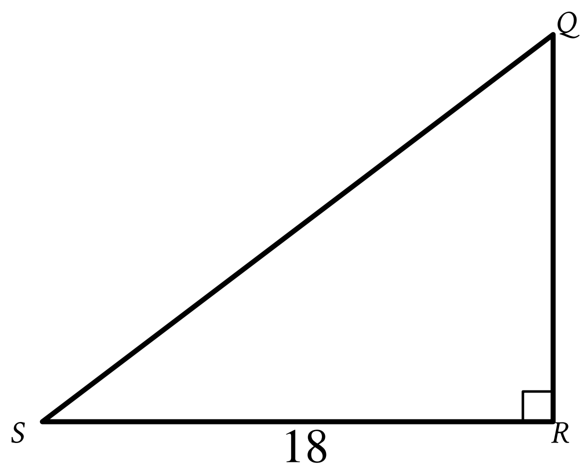
A circle in the xy-plane has its center at $(16, 17)$ and has a radius of $7k$. Which equation represents this circle?

A. $(x - 16)^2 + (y - 17)^2 = 49k$

B. $(x - 16)^2 + (y - 17)^2 = 49k^2$

C. $(x - 16)^2 + (y - 17)^2 = 7k$

D. $(x - 16)^2 + (y - 17)^2 = 7k^2$



Note: Figure not drawn to scale.

- Angle upper R is a right angle.
- The length of side upper R upper S is 18.
- A note indicates the figure is not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\bar{QS}$?

A. $18\cos Q$

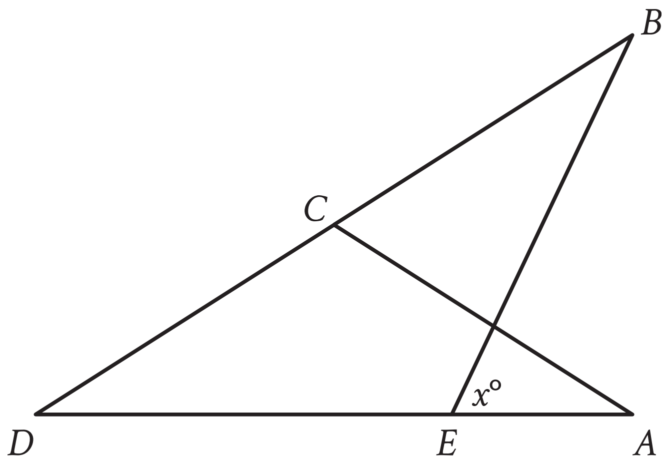
B. $18\sin Q$

C. $\frac{18}{\cos Q}$

D. $\frac{18}{\sin Q}$

Q7

GRID-IN GEOMETRY AND TRIGONOMETRY

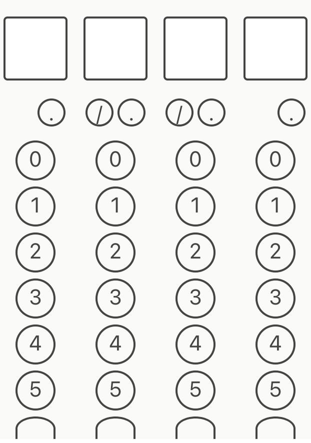


Note: Figure not drawn to scale.

- Triangle upper A upper C upper D partially overlaps triangle upper E upper B upper D.
- Point upper C is on line segment upper B upper D.
- Point upper E is on line segment upper A upper D.
- The measure of angle upper A upper E upper B is x°.
- A note indicates the figure is not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is $45°$, and the measure of angle ACD is $104°$. What is the value of x?

STUDENT-PRODUCED RESPONSE



6 6 6 6
7 7 7 7
8 8 8 8
9 9 9 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY

A right square prism has a height of 14 units. The volume of the prism is 2,016 cubic units. What is the length, in units, of an edge of the base?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GEOMETRY AND TRIGONOMETRY

A circle in the xy-plane has its center at $-1, 1$. Line t is tangent to this circle at the point $5, -4$. Which of the following points also lies on line t?

A. $0, \frac{6}{5}$

B. $4, 7$

C. $10, 2$

D. $11, 1$

Q10 GRID-IN GEOMETRY AND TRIGONOMETRY

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1{,}089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.